

## Training Configuration and Summary:

### Sequencing:

- timesteps: 30
- batch\_size: 128

### Model Hyperparameters:

- min\_units: 50
- max\_units: 150
- units\_step: 50
- dropout\_min: 0.2
- dropout\_max: 0.3
- dropout\_step: 0.1
- activation\_functions: ['tanh']
- optimizers: ['adam']

### Tuning:

- max\_epochs: 100
- hyperband\_iterations: 3

## BiGRU Multi-Step Training Report

Report Generated: 2025-05-03 19:12:21

Training Start: 2025-04-28\_12-57-33

Training End: 2025-04-28\_13-12-41

Training Duration: 15.13 minutes

### Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.1122

MSE: 1.8313

RMSE: 1.3533

R2 Score: 0.6974

### Multi-Step Forecast:

Forecast Steps: 10

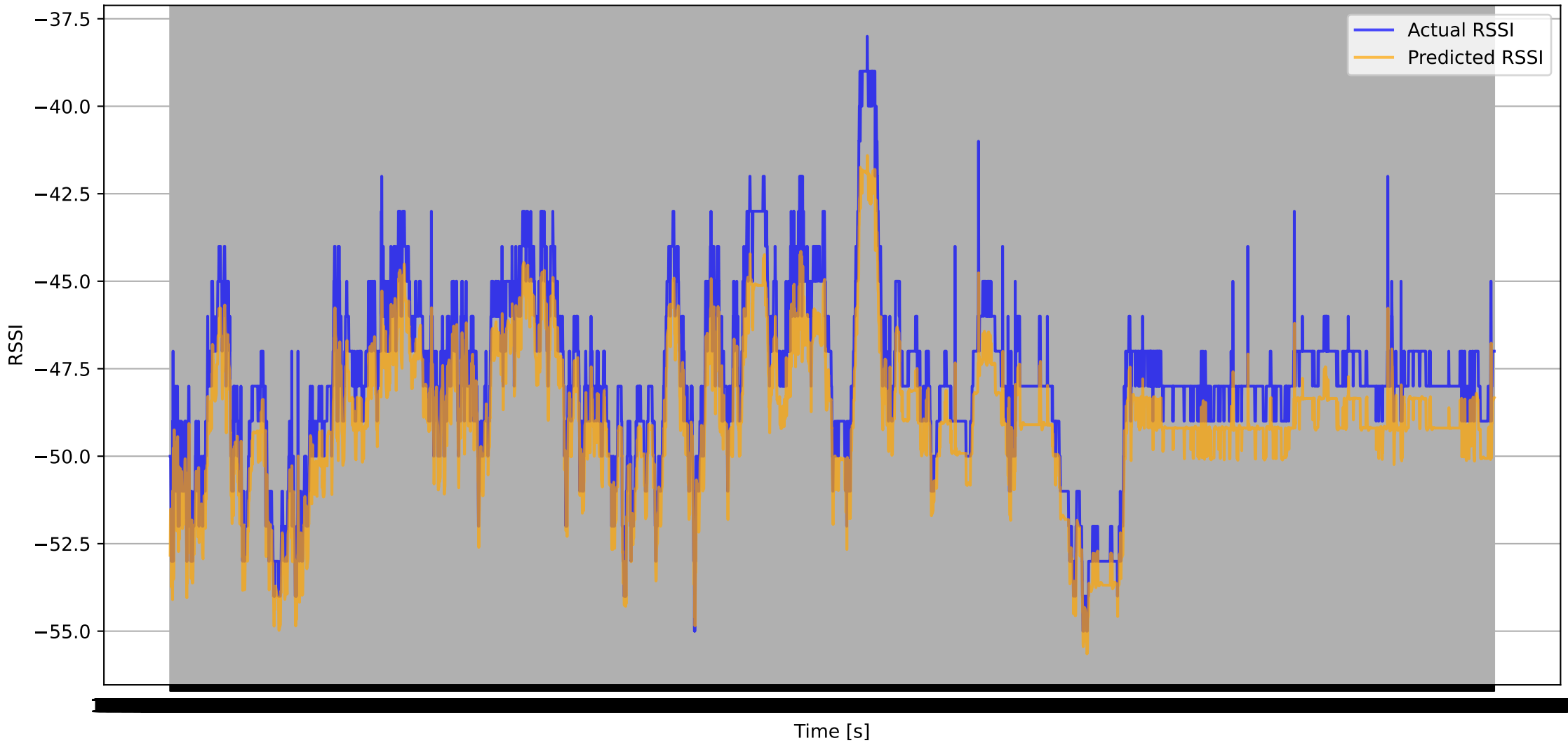
MAE: 0.0429

MSE: 0.0023

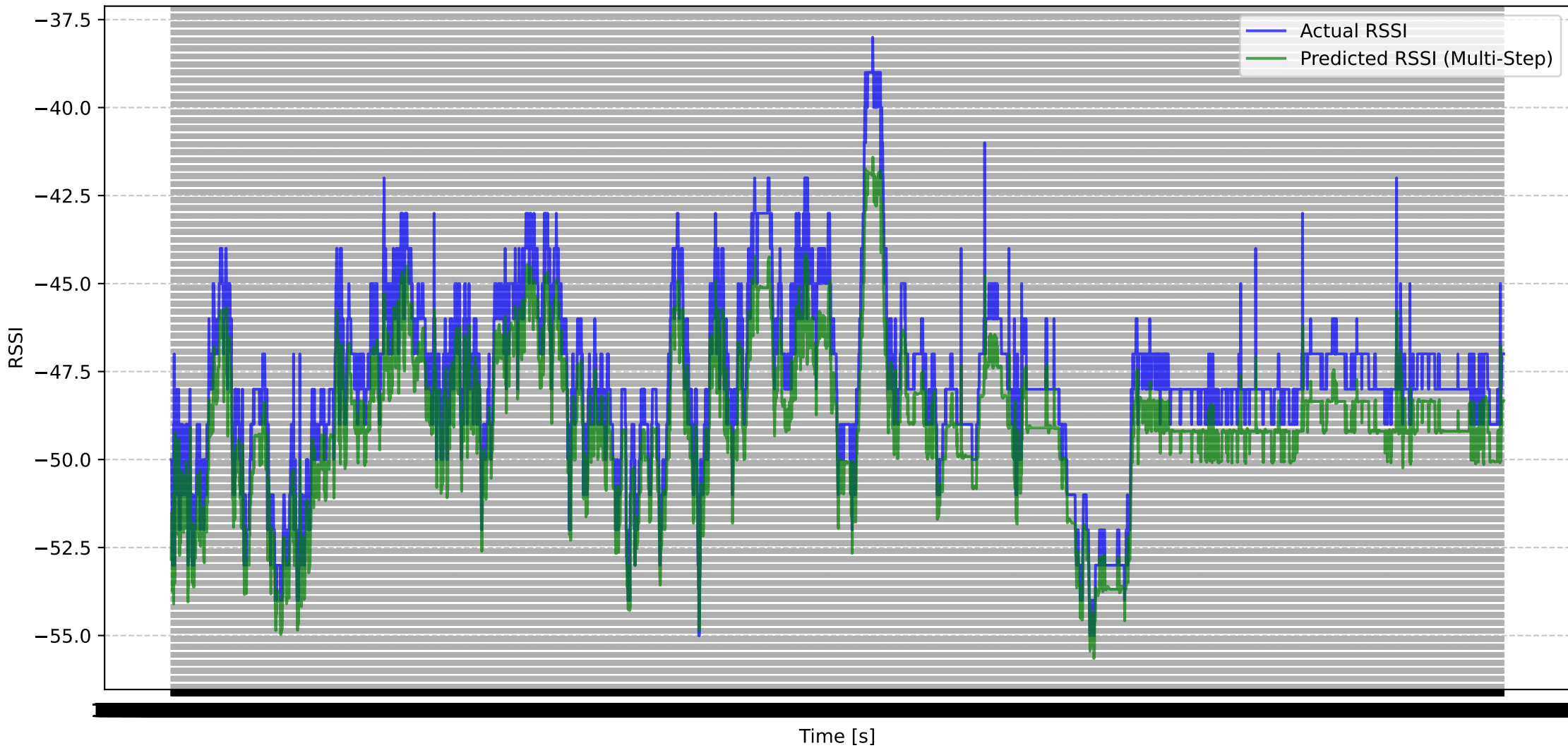
RMSE: 0.0476

R2 Score: 0.6858

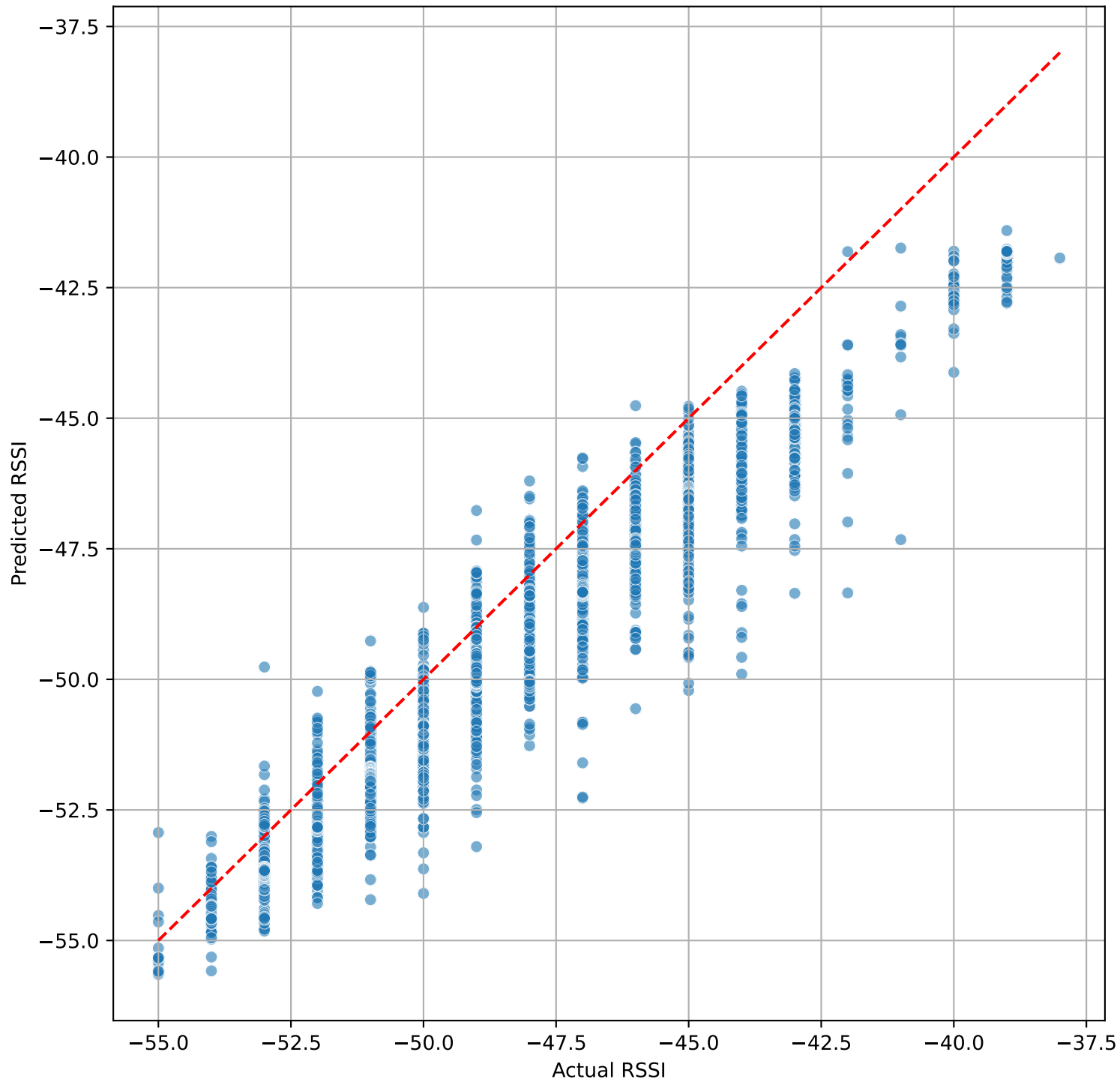
Actual vs Predicted RSSI (First Step)



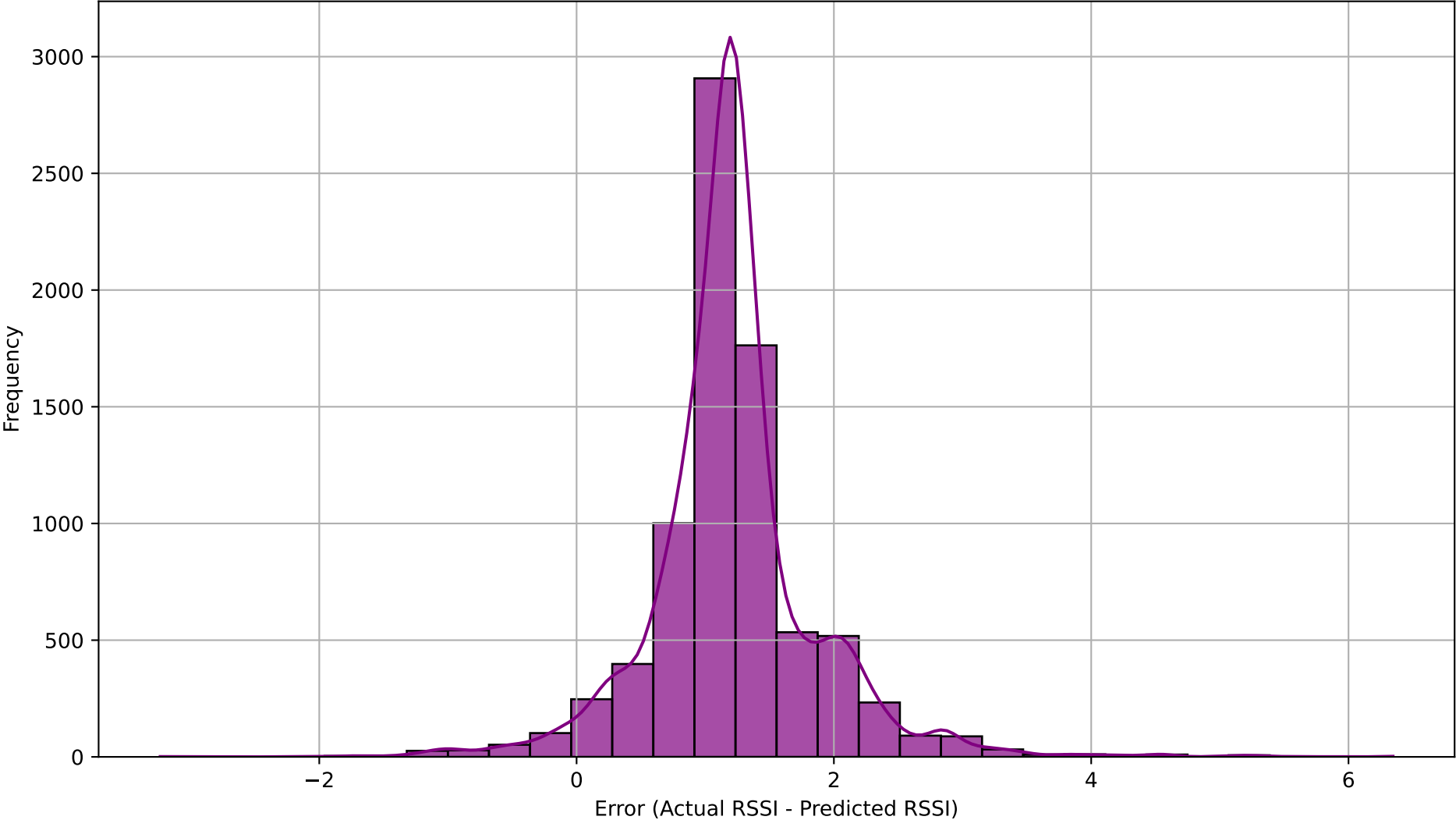
Actual vs Predicted RSSI (First Step vs Multi-Step)



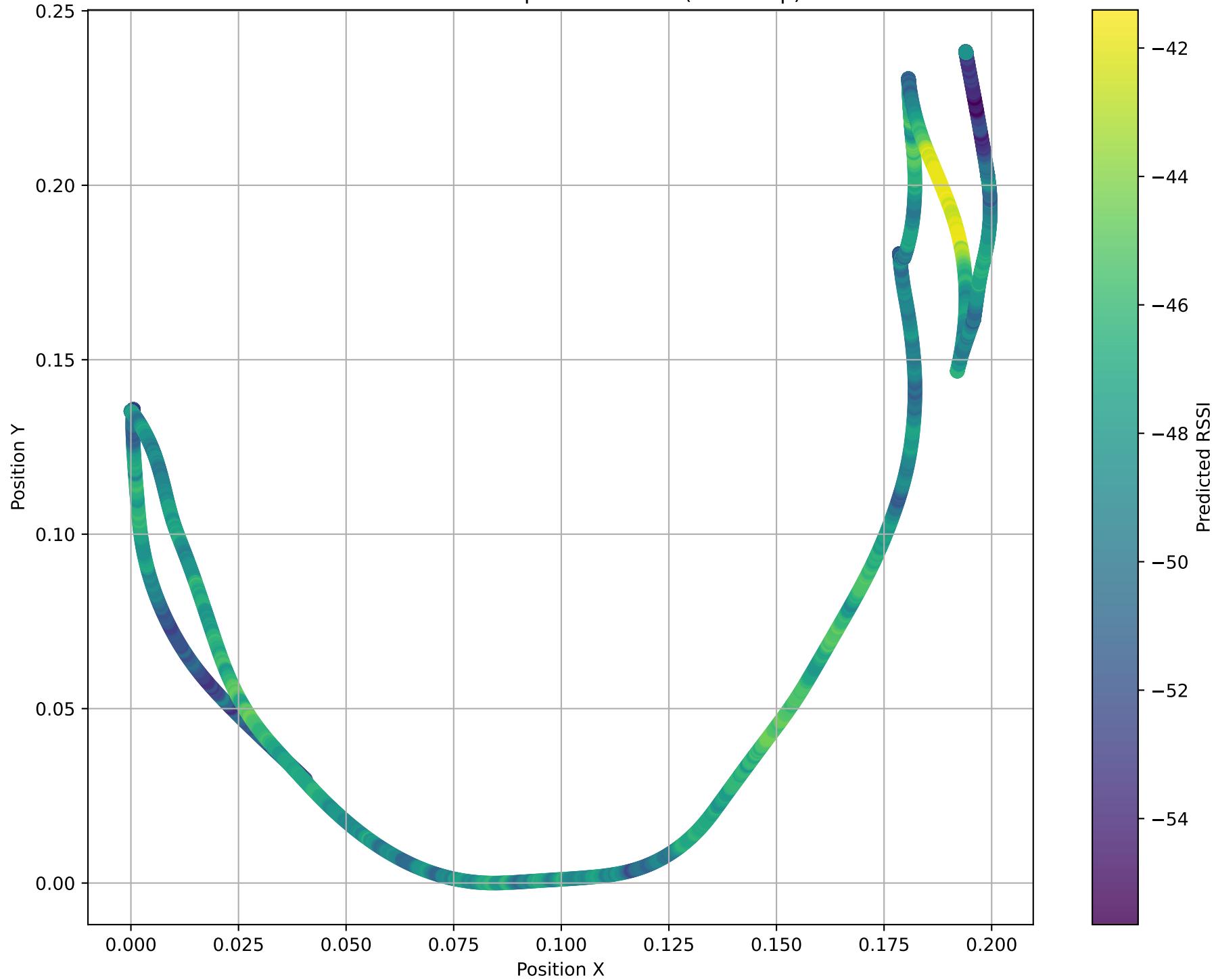
Actual vs Predicted RSSI (First Step)



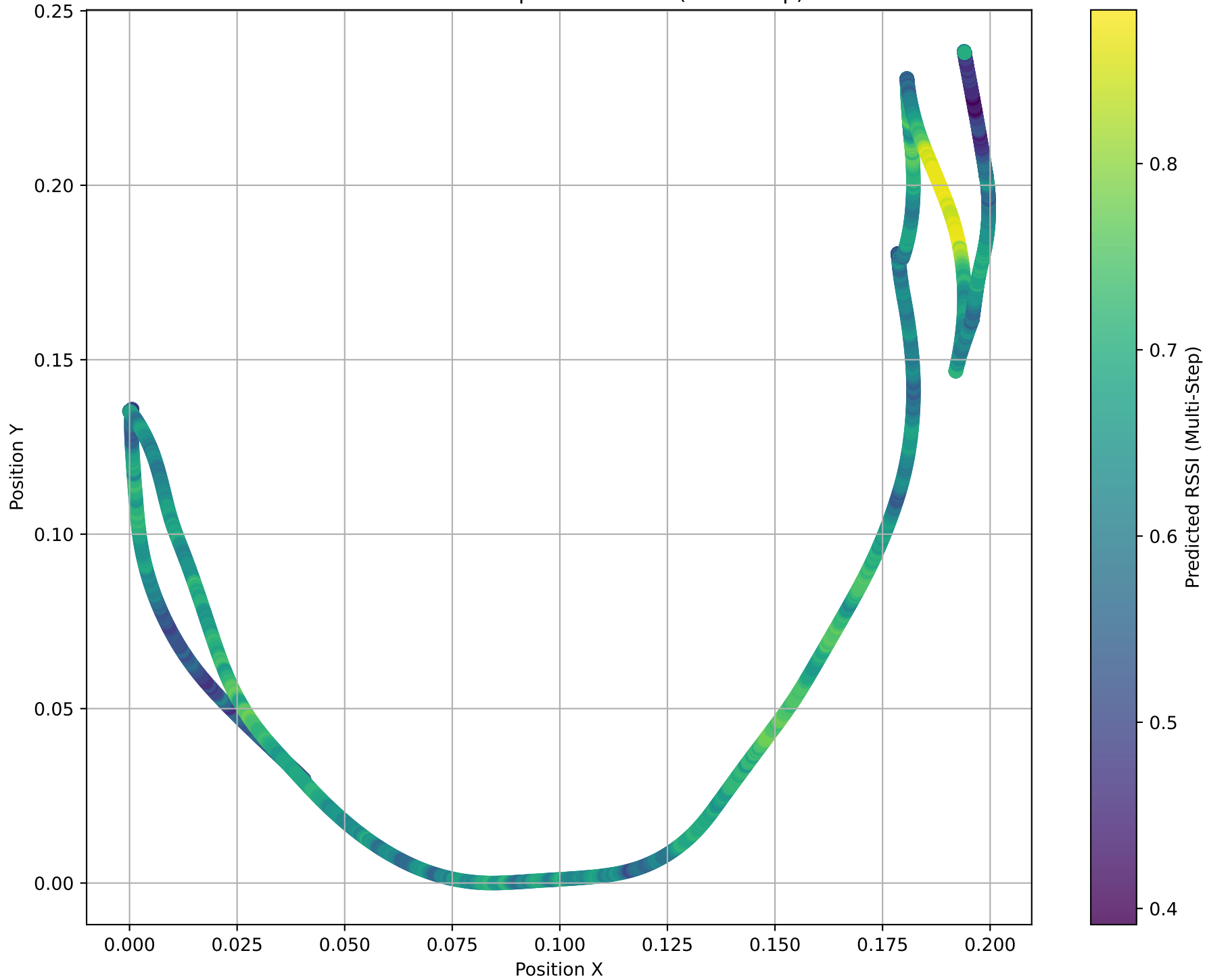
Distribution of Prediction Errors (First Step)



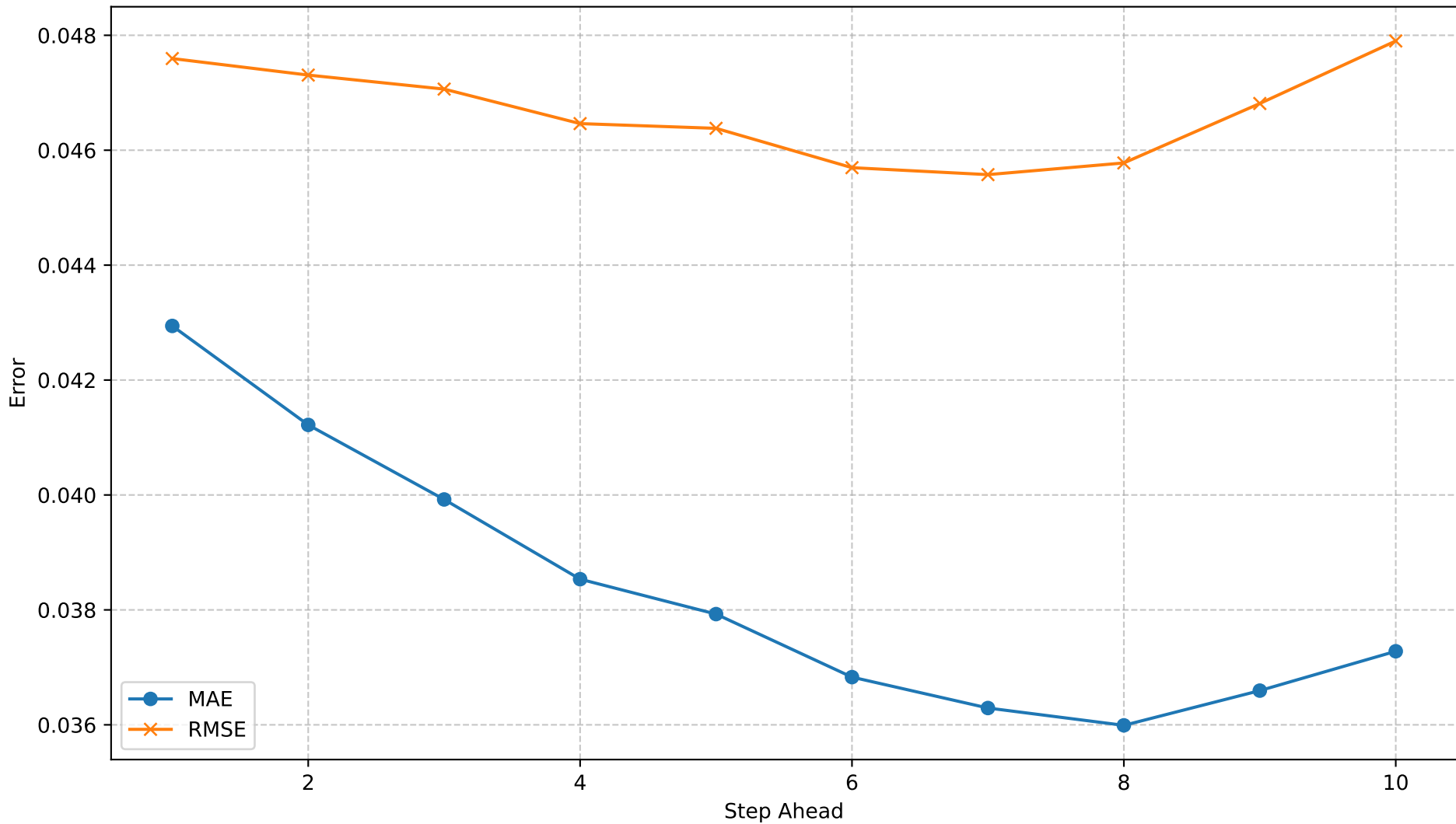
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step





#### Resumo Automático:

- Forecast Steps configurados: 10
- Passos com MAE  $\leq 2.0$ : 10 passo(s)
- MAE no último passo: 0.0373

[ V ] O modelo mantém boa acurácia durante quase toda a sequência de previsões.